Fe-C Problem 5 | Problems T4_PhaseDiagrams II.pdf #5 | Fe-C · lever rule

Composition from the Cementite Fraction

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

5.- Calculate the composition of a Fe-C alloy whose composition is within the range in %C of (2.11 < C_0 < 4.3). Consider temperature is (727–ΔT)ºC and the mass fraction χ_{Fe_3C} is 0.38.

Official Solution: C_0 = 2.56%C

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium; just below 727 °C every Fe-C alloy is α (0.022 % C) + Fe3C (6.67 % C) [Session 9 Slides 4, 9].
2. The range given (2.11-4.3 % C) is that of hypoeutectic cast irons (the diagram of the slides puts the steel limit at 2.14 %).
3. Unknown: C_0; one equation (lever rule for Fe3C).

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C), on the tie line at 727 – ΔT, from 0.022 to 6.67 % C; one-dimensional, no change of basis. The lever rule is solved for the alloy composition instead of the fraction [Session 7 Slide 16].

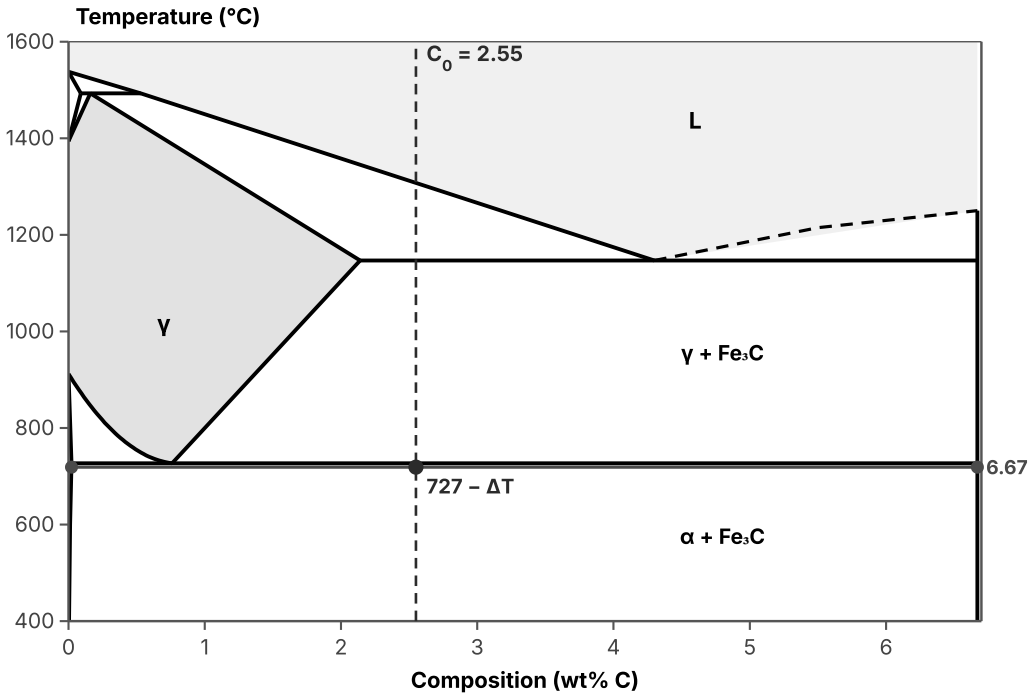


Figure 33. The alloy with 38 % Fe_3C just below 727 °C lies at 2.55 % C (computed in the problem).

PHASE 3 · STEP-BY-STEP DERIVATION

$$W_{\mathrm{Fe_3C}} = \frac{C_0 - C_\alpha}{C_{\mathrm{Fe_3C}} - C_\alpha} \;\Rightarrow\; C_0 = C_\alpha + W_{\mathrm{Fe_3C}}\,(C_{\mathrm{Fe_3C}} - C_\alpha)$$

$$C_0 = 0.022 + 0.38 \times (6.67 - 0.022) = 0.022 + 0.38 \times 6.648 = 0.022 + 2.526 = 2.55\ \%\,\mathrm{C}$$

PHASE 4 · CHECKS & INTERPRETATION

- Check: $(2.548 - 0.022)/6.648 = 0.380$; and $2.11 < 2.55 < 4.3$, inside the given range .
- Units: percentages of C on both sides; W dimensionless.
- The alloy is a hypoeutectic cast iron: it would contain primary austenite and ledeburite just below 1147 °C.

Check against the official solution

DISCREPANCY WITH THE OFFICIAL SOLUTION

The key gives $C_0 = 2.56$ % C, which follows from 6.7 % C for cementite: $0.022 + 0.38 \times 6.678 = 2.56$. With the 6.67 % C of the slides the result is 2.55 % C; the difference is only that rounding.